ERNESTINE HO

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EDUCATION

Toronto Metropolitan University (Formerly Ryerson University) ***Sep. 2020 – Jun. 2025***

Bachelor of Engineering – Computer Engineering

- Specialization in Software Engineering
- Dean's List Recipient – Fall 2024

SKILLS

Languages: Typescript/JavaScript, Python, Java, SQL, NoSQL, HTML/CSS, VBA, Kotlin

Frameworks: React, Next.js, Angular, Svelte, Node.js, FastAPI, Spring Boot

Developer Tools: Git, GitHub Copilot, AWS IoT, Azure, Azure DevOps, GitHub Actions, PostgreSQL, MySQL, MongoDB, OpenAI APIs, Docker, Kubernetes, gRPC, Appium, LambdaTest

EXPERIENCE

Software Developer Intern | Pomerleau ***Jan. 2024 – Sep. 2025***

- Revamped a real-time **React.ts (TypeScript)** interface integrated with **AWS IoT Core** and **Kinesis Video Streams**, achieving **5+** live feeds with **sub-2s latency**, while collaborating in an Agile environment.
- Managed Docker containers on **AWS IoT Greengrass**, exporting components and automating deployments through **AWS CodePipeline**, improving uptime by **40%**.
- Automated frontend testing with **Playwright**, reaching **95%** coverage, and reducing manual QA time by **50%**.

Software Developer Intern | LORA Technologies ***Jun. 2023 – Dec. 2023***

- Built a full-stack AI chatbot site using **Svelte**, **FastAPI**, and **OpenAI APIs** to answer financial questions.
- Automated verification of **100+** bot transactions using **VBA scripts** and running **SQL queries** in DataGrip against a PostgreSQL database, improving testing speed and reliability.
- Implemented automated validation workflows in the **Azure pipeline** using **Python** to verify PostgreSQL data quality and transactional dataset accuracy.
- Developed Java UI automation scripts using **Appium** and **LambdaTest**, reducing weekly downtime by over **4+ hours** and enhancing QA workflows through iterative problem-solving with the team.

Software Developer | Toronto Metrobotics ***Sep. 2021 – Jun. 2022***

- Developed a full-stack dashboard using **Spring Boot** and **React** with RESTful APIs and interactive charts, collaborating with the science team to translate requirements into actionable features supporting geological analysis.
- Developed a Tkinter dashboard to support real-time rover monitoring and maneuvering for the autonomous team.

PROJECTS

Social Networks-based Recommender System for Tourists (Capstone Project) ***Sep. 2024 – Apr. 2025***

- Developed a full-stack tourism recommender using **React.ts (Typescript)** and containerized **FastAPI** services with Docker, then deployed to **Azure Container Apps**.
- Built a **Databricks ETL pipeline** processing **3,800+ locations**, storing enriched results in **Azure Blob Storage**.
- Implemented a hybrid recommendation model with **OpenAI embeddings, scikit-learn** and **Surprise** models, improving **Top-10 accuracy by 18%**.

Horizon Chase - Personal Finance Banking Application ***Jan. 2026***

- Built a full-stack banking application using **Next.js 15** and **TypeScript** with **MongoDB** for scalable data persistence and **Better Auth** for secure user authentication.
- Designed responsive, accessible UI with **shadcn** component library and **Tailwind CSS**, following modern design system principles.
- Deployed application on **Vercel** with **Sentry** monitoring and automated CI/CD pipeline for production reliability.